

Week 2 Module

#1) Fraud Transaction Detection Date: .../.../...

Problem) A bank stores transaction IDs in the order they occur. Since the transactions are not sorted, the security team needs to check if a suspicious transaction ID exists.

To use:- Linear Search

Algorithm:-

Start-

Read input N.

Read no. of transaction IDs in array.

Read sus transaction ID.

Compare sus ID with every element.

If found print "Found".

If not print "Not Found".

Stop

Code)

```
import java.util.*; import java.util.Scanner;
public class LinearSearch {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];

        for (i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
    }
}
```

} Input array

```

for(int i=0; i<n; i++){
    if(arr[i] == target){
        found = true;
        break;
    }
}
if(found)
    System.out.println("Fraud Detected");
else
    s.println("not found");
}
}

```

Suppose, Input = 5
 1201 1305 1450 1502
 1450

Output;

~~Not~~

Fraud Detected

1201 1305 $\neq$ 1450(target) Not found

1450 = 1450 (target) Detected.

② Server log search

using

Binary
Search

```
Code: import java.util.Scanner;
public class BinarySearch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int key = sc.nextInt();
        int low = 0;
        int high = n - 1;
        boolean found = false;
        while (low <= high) {
            int mid = (low + high) / 2;

            if (arr[mid] == key) {
                found = true;
                break;
            }
            else if (arr[mid] < key)
                low = mid + 1;
            else
                high = mid - 1;
        }
        if (found)
            S.o.pln("log found");
        else
            S.o.pln("log not found");
    }
}
```

Dry Run

Date / /

Input: 7

(0)	(1)	(2)	(3)	(4)
101	105	109	102	115

Target = 109

Low = 0
high = 4 } index
no.

$$\text{Mid} = (0 + 4) / 2 = \cancel{20} 2$$

$$\text{arr}[2] = 109$$

Target Found

Output

Log Found.

Module 2

Update

LinearSearch.java BinarySearch.java BubbleSort.java InsertionSort.java SelectionSort.java

LinearSearch.java > LinearSearch > main(String[])

```
3 public class LinearSearch {
4     public static void main(String[] args) {
5
6
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9
10        for(int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13
14        int key = sc.nextInt();
15        boolean found = false;
16
17        for(int i = 0; i < n; i++) {
18            if(arr[i] == key) {
19                found = true;
20                break;
21            }
22        }
23
24        if(found)
25            System.out.println(x: "Fraud Transaction Found");
26        else
27            System.out.println(x: "Transaction Not Found");
28        sc.close();
29    }
30
31 }
32
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: LinearSearch

```
antimashukla@Antimas-MacBook-Air Module 2 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/antimashukla/Library/Application\ Support/Code/User/workspaceStorage/7c7c28e6216d085d1b84933848b186cb/redhat.java/jdt_ws/Module\ 2_3a24a8c1/bin LinearSearch
5
1201
1305
1450
1502
2074
1305
Fraud Transaction Found
antimashukla@Antimas-MacBook-Air Module 2 %
```

Ln 28, Col 16 Spaces: 4 UTF-8 LF {} Java Go Live

Week 2 Module

#1) Fraud Transaction Detection Date: .../.../...

Problem) A bank stores transaction IDs in the order they occur. Since the transactions are not sorted, the security team needs to check if a suspicious transaction ID exists.

To use:- Linear Search

Algorithm:-

Start-

Read input N.

Read no. of transaction IDs in array.

Read sus transaction ID.

Compare sus ID with every element.

If found print "Found".

If not print "Not Found".

Stop

Code)

```
import java.util.*; import java.util.Scanner;
public class LinearSearch {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];

        for (i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
```

} Input array

```

for(int i=0; i<n; i++){
    if(arr[i] == target){
        found = true;
        break;
    }
}
if(found)
    System.out.println("Fraud Detected");
else
    s.println("not found");
}
}

```

Suppose, Input = 5

1201 1305 1450 1502
1950

Output;

~~Not~~

Fraud Detected

1201 1305 $\neq$ 1450 (target) Not found

1950 = 1950 (target) Detected.

② Server log search

using

Binary Search

```
Code
import java.util.Scanner;
public class BinarySearch {
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int key = sc.nextInt();
        int low = 0;
        int high = n - 1;
        boolean found = false;
        while (low <= high) {
            int mid = (low + high) / 2;

            if (arr[mid] == key) {
                found = true;
                break;
            }
            else if (arr[mid] < key)
                low = mid + 1;
            else
                high = mid - 1;
        }
        if (found)
            S.o.pln("log found");
        else
            S.o.pln("log not found");
    }
}
```

Dry Run

Date / /

Input: 7

(0)	(1)	(2)	(3)	(4)
101	105	109	102	115

Target = 109

Low = 0
high = 4 } index
 no.

$$\text{Mid} = (0 + 4) / 2 = \cancel{20} 2$$

$$\text{arr}[2] = 109$$

Target Found

Output

Log. Found.

J SelectionSort.java

```

1 import java.util.Scanner;
2
3 public class BinarySearch {
4     Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9         int arr[] = new int[n];
10
11         for(int i = 0; i < n; i++) {
12             arr[i] = sc.nextInt();
13         }
14
15         int key = sc.nextInt();
16
17         int low = 0, high = n - 1;
18         boolean found = false;
19
20         while(low <= high) {
21             int mid = (low + high) / 2;
22
23             if(arr[mid] == key) {
24                 found = true;
25                 break;
26             }
27             else if(arr[mid] < key)
28                 low = mid + 1;
29             else
30                 high = mid - 1;
31         }
32     }
33 }

```

Run: BinarySearch

```
101
105
109
102
115
114
129
102
Log Found
antimashukla@Antimas-MacBook-Air Module 2 %
```

③ Sort Employee Salaries

Date .../.../.....

using) Bubble Sort

Read Input N $\rightarrow$ Read Salary Values $\rightarrow$ Compare element

Repeat until array gets sorted. $\leftarrow$ Swap if left is greater

Algorithm

- (1) Read Input in N.
- (2) Read Salary Values. in Array.
- (3) Compare array elements.
- (4) Swap when if left element is greater.
- (5) Repeat the step (4).
- (6) Stop.

Code

```
import java.util.Scanner;
public class BubbleSort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
    }
}
```

```
for(int i=0; i<n; i++) {
    S.O.Plm(arr[i] + " ");
}
```

3

Dry Run

Input 45 , 32 55 28 60

Pass 1) 45 > 32

Swap

32 45 55 28 60

Pass 2)

~~32~~ 45 < 55

Same

32 45 55 28 60

Pass 3)

55 > 28

swap

32 45 28 55 60

4)

55 < 60

Same

32 45 28 55 60

Repeat)

32 < 45 ✓, 45 > 28 swap

Dry Run.

Date .../.../.....

Input 32 45 55 28 60

pass 1) 32 < 45 no swap
45 < 55 no swap
~~28~~ 55 > 28, swap (left is greater)

32 45 28 55 60

Pass 2) 32 < 45, no swap
45 > 28, swap

32 28 45 55 60

Pass 3) 32 > 28, swap
28 32 45 55 60

therefore array is sorted.

Output)

28 32 45 55 60

LinearSearch.java BinarySearch.java BubbleSort.java X InsertionSort.java SelectionSort.java

BubbleSort.java > BubbleSort > main(String[])

```
3 public class BubbleSort {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9
10        for(int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13
14        for(int i = 0; i < n - 1; i++) {
15            for(int j = 0; j < n - i - 1; j++) {
16                if(arr[j] > arr[j + 1]) {
17                    int temp = arr[j];
18                    arr[j] = arr[j + 1];
19                    arr[j + 1] = temp;
20                }
21            }
22        }
23
24        for(int i = 0; i < n; i++) {
25            System.out.print(arr[i] + " ");
26        }
27        sc.close();
28    }
29
30 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
antimashukla@Antimas-MacBook-Air Module 2 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/antimashukla/Library/Application\ Support/Code/User/workspaceStorage/7c7c28e6216d085d1b84933648b186cb/redhat.java/jdt_ws/Module\ 2_3a24a8c1/bin BubbleSort
4
32
45
55
28
28 32 45 55
antimashukla@Antimas-MacBook-Air Module 2 %
```

Run: Binary...

Run: Bubbl...

④ To sort website
response time

Date / /

Using Insertion sort:-

Insert each element in its correct position
in sorted position.

Ex) If $arr = [25, 12, 32, 18, 10]$

then) second element = key element = 12

$25 > 12$, 12 gets shifted

12 25 32 18 10

acc. to loop, key = 32 (arr[2])

32 > 25, no change

now, 18 < 32 (shift)

12 18 25 32 10

then, key = 10

10 12 18 25 32

Code)

Date / ... /

```
import java.util.Scanner;
```

```
public class InsertionSort {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int arr[] = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 1; i < n; i++) {
```

```
            int key = arr[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && arr[j] > key) {
```

```
                arr[j+1] = arr[j];
```

```
                j--;
```

```
            } arr[j+1] = key;
```

```
        }
```

```
        for (int i = 0; i < n; i++) {
```

```
            System.out.print(arr[i] + " ");
```

```
        }
```

```
    }
```

```
}
```

LinearSearch.java BinarySearch.java BubbleSort.java InsertionSort.java X SelectionSort.java

▶ 📄 ...

InsertionSort.java > InsertionSort > main(String[])

```
1  import java.util.Scanner;
2
3  public class InsertionSort {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          int n = sc.nextInt();
8          int arr[] = new int[n];
9
10         for(int i = 0; i < n; i++) {
11             arr[i] = sc.nextInt();
12         }
13
14         for(int i = 1; i < n; i++) {
15             int key = arr[i];
16             int j = i - 1;
17
18             while(j >= 0 && arr[j] > key) {
19                 arr[j + 1] = arr[j];
20                 j--;
21             }
22
23             arr[j + 1] = key;
24         }
25         for(int i = 0; i < n; i++) {
26             System.out.print(arr[i] + " ");
27         }
28         sc.close();
29     }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
antimashukla@Antimas-MacBook-Air Module 2 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/antimashukla/Library/Application\ Support/Code/User/workspaceStorage/7c7c28e6216d085d1b84933648b186cb/redhat.java/jdt_ws/Module\ 2_3a24a8c1/bin InsertionSort
5
25
12
32
18
10
10 12 18 25 32
antimashukla@Antimas-MacBook-Air Module 2 %
```

Run: Binary...

Run: Bubbl...

Run: Inserti...

⑤ To prioritize Bug Reports

(Selection Sort) Date / /

In Selection Sort,

it repeatedly finds smallest element and places it at its correct position.

Suppose, arr[i] = 9 2 8 ① 5
smallest

Pass 1 $\Rightarrow$ ① 9 2 8 5
then,

Pass 2 $\Rightarrow$ ① ② 9 8 5

Pass 3 $\Rightarrow$ ① ② ③ 9 8

Pass 4 $\Rightarrow$ ① ② ③ ④ ⑤
sorted

Code

```
import java.util.Scanner;  
public class SelectionSort {  
    public static void main (String args[]) {
```

```
        Scanner sc = new Scanner (System.in);  
        int n = sc.nextInt();  
        int arr[] = new int[n];
```

```
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
        for (int i = 0; i < n - 1; i++) {  
            int min = i;
```

J LinearSearch.java J BinarySearch.java J BubbleSort.java J InsertionSort.java J SelectionSort.java 3

J SelectionSort.java > SelectionSort > main(String[])

```
3 public class SelectionSort {
4 public static void main(String[] args) {
9
10     for(int i = 0; i < n; i++) {
11         arr[i] = sc.nextInt();
12     }
13
14     for(int i = 0; i < n - 1; i++) {
15         int min = i;
16
17         for(int j = i + 1; j < n; j++) {
18             if(arr[j] < arr[min]) {
19                 min = j;
20             }
21         }
22
23         int temp = arr[i];
24         arr[i] = arr[min];
25         arr[min] = temp;
26     }
27
28     for(int i = 0; i < n; i++) {
29         System.out.print(arr[i] + " ");
30     }
31     sc.close();
32 }
33
34 }
35
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
antimashukla@Antimas-MacBook-Air Module 2 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/antimashukla/Library/Application\ Support/Code/User/workspaceStorage/7c7c28e6216d085d1b84933648b186cb/redhat.java/jdt_ws/Module\ 2_3a24a8c1/bin SelectionSort
5
9
2
8
1
5
1 2 5 8 9
antimashukla@Antimas-MacBook-Air Module 2 %
```

Run: Binary...
Run: Bubbl...
Run: Inserti...
Run: Select...